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1. Executive Summary

The Bishoftu Guest House Management System (GHMS) with Police Integration is a transformative digital platform designed to modernize the guest house industry across Bishoftu City and its three sub-city administrations: Cheleleka, Dibayyu, and Dukem. This proposal presents a comprehensive, technology-driven solution that addresses the critical gap between hospitality service management and public safety enforcement. The system digitizes guest house operations including reservations, room management, guest registration, payment tracking, and expense management, while simultaneously providing law enforcement agencies with real-time intelligence capabilities for suspect identification, geofence monitoring, and cross-referencing of guest records against wanted persons databases.

Currently, guest houses in the Bishoftu area operate through manual, paper-based processes that create significant vulnerabilities in both service quality and security. Guest registration records exist in isolated ledgers, police verification depends entirely on phone calls and physical visits, and there is no centralized mechanism for tracking guest movement across multiple establishments. This fragmentation has created an environment where suspicious activities can go undetected, regulatory compliance cannot be effectively monitored, and the hospitality sector cannot reach its full economic potential. The proposed system eliminates these gaps by creating a unified digital ecosystem that serves all stakeholders simultaneously: guest house owners gain professional management tools, guests gain transparency and direct access to services, police gain real-time intelligence, and city administrators gain comprehensive oversight and regulatory enforcement capabilities.

The platform is built on modern web technologies including Next.js 16, PostgreSQL, Prisma ORM, and Vercel cloud infrastructure, ensuring scalability, security, and reliability. A guest-facing Progressive Web Application (PWA) enables citizens to browse registered guest houses, view room availability, and report security concerns directly to law enforcement. The system has been designed with role-based access control supporting five distinct user roles: Superuser (system administrator), Operator (city administration), Police (law enforcement with ranked hierarchy), Staff (guest house employees), and Guests (public users). This multi-stakeholder architecture ensures that every party in the ecosystem has the precise access and tools they need, while maintaining strict data isolation between sensitive police intelligence and routine hospitality operations.

2. Problem Statement: Drawbacks of the Existing System

The current hospitality management landscape in Bishoftu City suffers from systemic inefficiencies that affect public safety, economic development, and regulatory compliance. Through extensive consultation with guest house operators, local police officials, and city administration representatives, the following critical drawbacks have been identified in the existing manual system. Each of these issues represents not merely an inconvenience but a structural failure that undermines the safety and prosperity of the community.

2.1 Paper-Based Guest Registration

Guest houses across Bishoftu currently rely on handwritten registration books to record guest information. These paper records are inherently vulnerable to damage, loss, and forgery. More critically, they are completely isolated from one another, meaning a guest who moves between multiple establishments leaves no connected trail. When police need to investigate a person of interest, they must physically visit each guest house individually and manually search through paper ledgers, a process that can take days or weeks and often yields incomplete results. The absence of a centralized digital database means that patterns of suspicious behavior, such as a single individual frequently changing locations or using different identification documents, are virtually impossible to detect under the current system.

2.2 No Real-Time Police Integration

There is currently no digital communication channel between guest houses and law enforcement agencies. Police officers must rely on informal relationships and physical visits to gather intelligence about guest activities. This creates dangerous delays in identifying and apprehending suspects. If a wanted criminal checks into a guest house, the establishment has no automated mechanism to flag this check-in to authorities. Similarly, if a guest reports suspicious activity, the report must travel through informal channels rather than being instantly routed to the nearest police station with precise location data and guest details. This lack of real-time integration has been identified by law enforcement as one of the most significant security gaps in the current hospitality oversight framework.

2.3 Absence of Regulatory Oversight

The city administration has no systematic way to track which guest houses are operating legally, whether they meet minimum standards, or whether they have renewed their operating licenses. Guest houses can operate for years without any inspection or compliance verification. There is no centralized registry of licensed establishments, no mechanism for tracking license renewals, and no standardized criteria for evaluating operational quality. This regulatory vacuum not only exposes guests to substandard and potentially unsafe accommodations but also creates opportunities for unlicensed establishments to operate outside the law, potentially harboring illegal activities without any risk of detection or consequence.

2.4 No Guest-Facing Service Discovery

Visitors and tourists arriving in Bishoftu have no digital platform to discover available guest houses, compare room prices, or verify the legitimacy of an establishment. This information asymmetry forces guests to rely on word-of-mouth recommendations or physical visits, which is particularly challenging for travelers arriving from other cities or countries. The absence of a public-facing platform also means guests have no channel to report problems such as safety concerns, overcharging, or fraudulent practices to the relevant authorities. This lack of transparency undermines consumer confidence in the local hospitality sector and limits the economic growth potential of the industry, as visitors cannot make informed choices about their accommodation.

2.5 Manual Financial and Operational Management

Guest house operators manage their businesses using a combination of paper ledgers, personal spreadsheets, and memory. Room availability is tracked manually on whiteboards or in notebooks, leading to frequent double-bookings and lost revenue. Financial records including payments, expenses, and tax obligations are maintained in informal systems that make accurate reporting impossible. The lack of professional management tools means guest house owners cannot analyze their business performance, identify trends, or make data-driven decisions about pricing, staffing, or facility improvements. This operational inefficiency directly limits the profitability and sustainability of guest house businesses in the region.

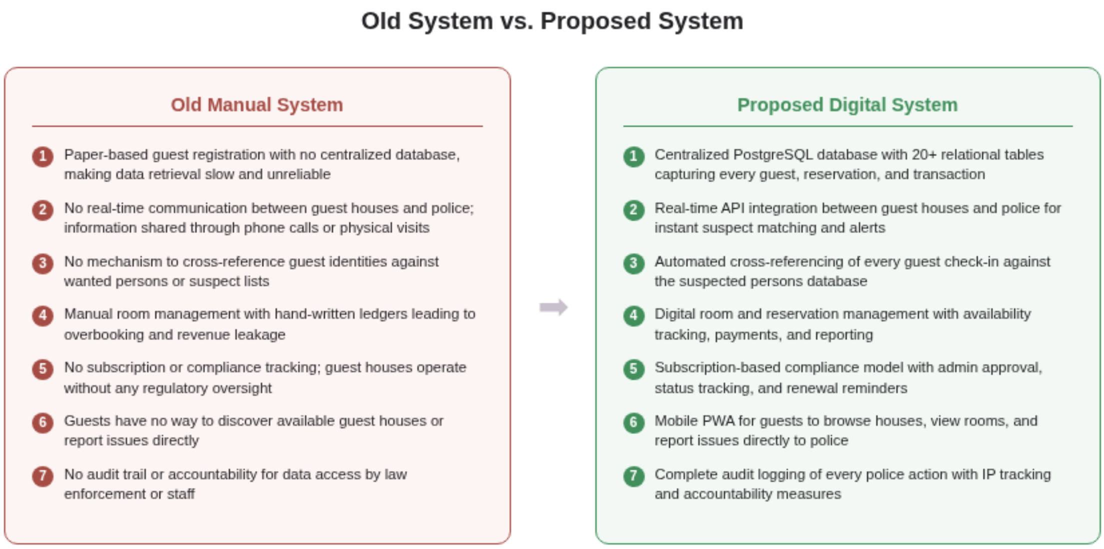

Figure 1: Comparison of Old Manual System vs. Proposed Digital System

3. Proposed Solution

The proposed Bishoftu Guest House Management System is a comprehensive, cloud-hosted digital platform that addresses every identified drawback of the current manual system. The solution is designed as a multi-tenant web application with five distinct interfaces tailored to the specific needs of each stakeholder group. Built on a modern technology stack, the system ensures reliability, security, and scalability while remaining accessible to users with varying levels of technical expertise. The following sections describe the core components and capabilities of the proposed system in detail.

3.1 System Architecture

The system follows a four-layer architecture that ensures clean separation of concerns, maintainability, and security. The Presentation Layer consists of four distinct user interfaces: a mobile Progressive Web App for guests, a comprehensive admin dashboard for superusers and operators, a dedicated police intelligence portal, and a staff management dashboard for guest house employees. Each interface is designed with mobile-first principles to ensure accessibility across all device types. The API Gateway Layer is built on Next.js server-side route handlers and provides RESTful endpoints for all system operations, with public endpoints for guest-facing features and authenticated endpoints for administrative and police functions. The Intelligence and Security Engine operates as a middleware layer that performs automated suspect matching, geofence monitoring, anomaly detection, and comprehensive audit logging. The Data Layer uses PostgreSQL as the primary database with Prisma ORM for type-safe data access, managed through 20+ relational tables covering every aspect of guest house operations and police intelligence. The system is deployed on Vercel cloud infrastructure for automatic scaling, global edge deployment, and continuous integration and delivery.

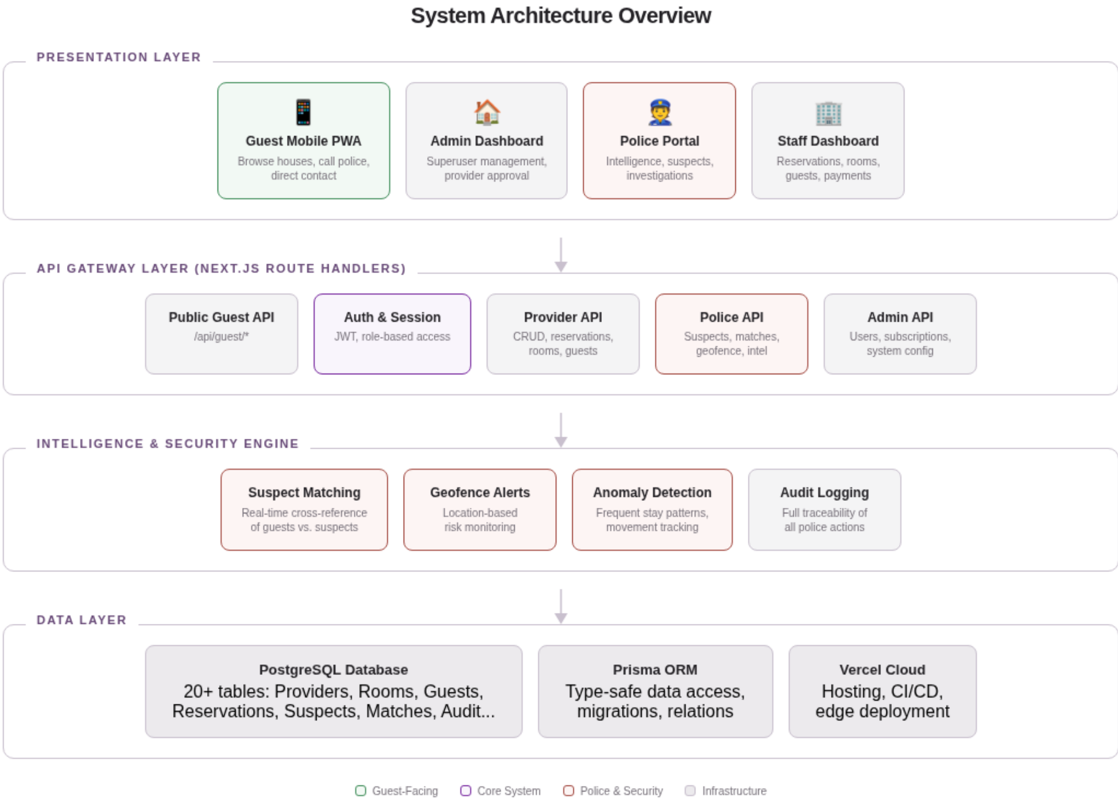

Figure 2: System Architecture Overview - Four-Layer Design

3.2 Guest House Management Module

The guest house management module provides a complete digital solution for daily operations. Room management supports five room types (Single, Double, Twin, Suite, and Deluxe) with real-time availability tracking across four status states (Available, Occupied, Maintenance, and Reserved). The reservation system handles the full lifecycle from booking creation through check-in, check-out, and cancellation, with support for multiple payment methods including cash, bank transfer, card, and mobile money. Guest registration captures comprehensive information including identification documents, nationality, and full address details broken down by region, zone, woreda, and kebele for precise geographic tracking. Financial management includes expense tracking with categorized entries, payment recording with receipt numbers, and tax calculation support. Housekeeping task management enables scheduling of cleaning, maintenance, and inspection tasks with assignment tracking and completion status monitoring. The module also supports daytime service bookings for hourly services such as conference room usage, laundry, and dining, providing guest houses with additional revenue stream management.

3.3 Police Intelligence and Security Module

The police intelligence module represents the most significant innovation in this system, providing law enforcement with capabilities that were previously impossible under the manual system. The Suspected

Persons database allows police to maintain a comprehensive registry of persons of interest with severity ratings (Low, Medium, High, Critical), identification documents, and behavioral descriptions. When a new guest checks into any registered guest house, the system automatically cross-references the guest information against this database and generates real-time match alerts when potential matches are detected. The Geofence feature enables police to define virtual perimeters around sensitive locations, with automatic alerts triggered when registered guests enter or exit these zones. The Frequent Stay Analysis engine identifies patterns such as guests who stay at unusually short intervals across multiple establishments, a behavior pattern that may indicate surveillance, drug trafficking, or other illicit activities. All police actions within the system are recorded in a comprehensive Audit Log that tracks the officer name, action type, target, IP address, and timestamp, ensuring full accountability and traceability of law enforcement access to sensitive guest data.

Feature	Description	Impact
Suspect Matching	Auto cross-reference every check-in against suspect database	Real-time threat detection
Geofence Alerts	Virtual perimeters with automatic entry/exit notifications	Location-based security monitoring
Frequent Stay Analysis	Pattern detection for guests moving between establishments	Early warning for suspicious behavior
Audit Logging	Full traceability of all police data access and actions	Accountability and transparency
Data Export	Structured intelligence reports for investigations	Evidence-grade documentation
Provider Suspension	Emergency closure authority for non-compliant houses	Rapid regulatory enforcement

Table 1: Police Intelligence and Security Module Features

3.4 Guest-Facing Mobile Application (PWA)

The guest-facing Progressive Web Application provides citizens with a modern, mobile-optimized interface for discovering and engaging with registered guest houses in the Bishoftu area. The PWA features a searchable directory of all approved guest houses organized by sub-city (Cheleleka, Dibayyu, and Dukem), allowing visitors to filter by location and search by name. Each guest house listing displays the property name, location details, available room count, and minimum nightly rate. The application includes a prominent "Call Police" button that connects guests directly to the Bishoftu police emergency line, ensuring that any security concern can be reported immediately. The PWA is designed as an installable application that can be added to a phone home screen for instant access, functioning like a native app without requiring installation from an app store. This approach ensures maximum accessibility across all smartphone types and eliminates the barrier of app store discovery and download requirements.

3.5 Administrative and Compliance Module

The administrative module provides city administrators (Superusers and Operators) with comprehensive oversight tools for managing the entire guest house ecosystem. Provider registration requires submission of business license information, physical address broken down by administrative hierarchy (sub-city, woreda), and contact details. All new registrations undergo administrative review before activation, ensuring that only legitimate, licensed establishments appear in the public system. The subscription management system supports four billing cycles (Monthly, Quarterly, Semi-Annual, and Yearly) with automated payment tracking and renewal notifications. Administrators can view comprehensive dashboards showing registration statistics, subscription compliance rates, and system-wide operational metrics. The system also supports joint operations between police and city administration through shared emergency response tools and unified user management for cross-agency collaboration.

4. System Visibility and Project Impact

The visibility of this project extends across multiple dimensions, encompassing public safety improvements, economic development, regulatory modernization, and technological advancement. This section outlines the tangible and measurable impacts that the system will deliver to each stakeholder group and the broader Bishoftu community. The project is designed not merely as a software deployment but as a foundational infrastructure investment that will transform how the hospitality sector operates and interacts with law enforcement for years to come.

4.1 Public Safety Enhancement

The most immediate and impactful benefit of this system is the dramatic enhancement of public safety through real-time intelligence sharing between guest houses and police. The automated suspect matching system will reduce the time required to identify a wanted person checking into a guest house from days or weeks under the current manual system to seconds. The geofence monitoring capability provides continuous, automated surveillance of sensitive areas without requiring additional police personnel. The frequent stay analysis engine will enable proactive identification of suspicious patterns before crimes occur, shifting the security posture from reactive investigation to proactive prevention. The direct police communication channel in the guest mobile application empowers citizens to report security concerns instantly, creating a community-wide safety network that extends the reach of law enforcement beyond what traditional policing methods can achieve.

4.2 Economic Development

The system will drive economic growth in the Bishoftu hospitality sector through several mechanisms. The public guest-facing platform will increase visibility for registered guest houses, connecting them with potential customers who previously had no way to discover their services. Professional management tools will help guest house operators improve operational efficiency, reduce double-bookings, and optimize room pricing, directly increasing revenue per available room. The subscription-based compliance model creates a sustainable revenue stream for system maintenance while ensuring that only serious, committed operators participate in the formal economy. As the system demonstrates its effectiveness in improving both safety and service quality, Bishoftu will gain a reputation as a well-regulated, safe destination for both domestic and international visitors, attracting additional tourism revenue and encouraging further investment in the hospitality sector.

4.3 Regulatory Modernization

For city administrators, this system provides the first comprehensive digital tool for regulating the guest house industry. The centralized provider registry gives administrators real-time visibility into which establishments are operating, their license status, and their compliance history. Subscription tracking automates what was previously a manual and unreliable process of tracking license renewals and fee payments. The dashboard analytics provide data-driven insights into industry trends, enabling evidence-based policy decisions. The system creates an auditable record of all administrative actions, ensuring transparency and accountability in regulatory enforcement. This digital transformation of regulatory processes sets a precedent that can be extended to other sectors of city administration, establishing Bishoftu as a leader in digital governance within the Oromia region.

4.4 Key Performance Indicators

The success of this project will be measured against the following key performance indicators, which have been designed to provide clear, quantifiable metrics for evaluating the system impact across all stakeholder groups. These KPIs will be tracked through the system built-in analytics dashboard and reported to city administration on a quarterly basis.

KPI	Current Baseline	Target (Year 1)	Target (Year 2)
Guest house registration rate	Unknown (no tracking)	80% of operating houses	95% coverage

KPI	Current Baseline	Target (Year 1)	Target (Year 2)
Suspect identification time	Days to weeks (manual)	Under 5 minutes (automated)	Real-time (instant)
Guest satisfaction score	Not measured	3.5 / 5.0 average	4.2 / 5.0 average
Revenue per room (avg increase)	Not tracked	15% improvement	25% improvement
Police response time	Hours (physical visit)	Minutes (digital alert)	Minutes (direct PWA call)
Compliance rate (licenses)	Estimated below 50%	75% compliance	90% compliance
Digital check-in adoption	0% (all paper)	60% digital registrations	85% digital registrations

Table 2: Key Performance Indicators and Targets

5. Technology Stack

The system is built on a carefully selected technology stack that prioritizes reliability, security, scalability, and maintainability. Each technology choice has been made to ensure that the system can serve the Bishoftu community effectively while being maintainable by local developers and sustainable within the available infrastructure. The following table summarizes the core technologies and their roles within the system.

Component	Technology	Purpose
Frontend Framework	Next.js 16 + React	Full-stack web application with server-side rendering
UI Components	Tailwind CSS + shadcn/ui	Professional, responsive interface design
Database	PostgreSQL	Relational data storage with advanced querying
ORM	Prisma	Type-safe database access and migration management
Authentication	JWT + bcrypt	Secure role-based access control
Hosting	Vercel Cloud	Automatic scaling, CI/CD, global edge deployment
Guest App	Progressive Web App (PWA)	Installable mobile experience without app store

Component	Technology	Purpose
Real-time Alerts	REST API + Web hooks	Instant suspect matching and geofence notifications

Table 3: Technology Stack Summary

6. Implementation Roadmap

The implementation of this system will follow a phased approach designed to minimize disruption to ongoing operations while delivering incremental value at each stage. The roadmap has been designed in consultation with stakeholders to ensure that the most critical capabilities are deployed first, building momentum and demonstrating value before proceeding to more advanced features.

Phase	Duration	Deliverables	Success Criteria
Phase 1: Foundation	Months 1-3	Core platform, provider registration, admin dashboard, basic room and reservation management	10+ guest houses registered and actively using the system
Phase 2: Security	Months 3-5	Police intelligence module, suspect database, automated matching, audit logging	Police actively using the system for suspect verification
Phase 3: Public Access	Months 5-7	Guest PWA launch, public directory, direct police communication, sub-city filtering	PWA installed by 100+ users within first month
Phase 4: Intelligence	Months 7-9	Geofence monitoring, frequent stay analysis, anomaly detection, data export tools	At least 2 geofence zones actively monitored
Phase 5: Scale	Months 9-12	Subscription management, analytics dashboards, performance optimization, city-wide expansion	All sub-cities (Cheleleka, Dibayyu, Dukem) fully onboarded

Table 4: Implementation Roadmap and Phased Delivery Plan

7. Budget Estimate

The following budget estimate provides a high-level overview of the investment required to implement and operate the Bishoftu Guest House Management System. The estimates are based on current market rates for cloud infrastructure, software development, and training services within Ethiopia. The budget is structured to demonstrate that a high-impact digital platform can be deployed within a reasonable investment envelope, with the subscription model providing a path to long-term financial sustainability that reduces dependence on continued government funding.

Category	Description	Estimated Cost (ETB)
Platform Development	Full-stack development, testing, and deployment of all modules	1,500,000
Cloud Infrastructure (Year 1)	Vercel hosting, PostgreSQL database, domain, SSL certificates	120,000
Training and Onboarding	Staff training for guest houses, police, and administrators	200,000
PWA Design and Testing	Mobile interface design, usability testing, icon and asset creation	150,000
Project Management	Coordination, stakeholder engagement, progress monitoring	180,000
Contingency (15%)	Risk buffer for unforeseen requirements and scope adjustments	322,500
Total Investment	Complete first-year implementation and operational costs	2,472,500

Table 5: Budget Estimate for First-Year Implementation

The projected annual operational cost from Year 2 onward is estimated at approximately 250,000 ETB, covering cloud hosting, maintenance, and support. This operational cost can be fully offset by the subscription revenue model, where each registered guest house pays a monthly or quarterly subscription fee. With a target of 50 registered guest houses paying an average of 500 ETB per month, the projected annual subscription revenue would be 300,000 ETB, making the system financially self-sustaining from the second year of operation.

8. Conclusion and Recommendation

The Bishoftu Guest House Management System with Police Integration represents a critical investment in the safety, economic development, and regulatory modernization of the Bishoftu City hospitality sector. The current manual system creates unacceptable gaps in public safety, limits economic growth, and prevents effective regulatory oversight. The proposed digital platform addresses all of these challenges through a single, integrated solution that serves every stakeholder in the ecosystem: guest house operators gain professional management tools, guests gain transparency and safety, police gain real-time intelligence capabilities, and city administrators gain comprehensive oversight and regulatory enforcement tools.

The technology stack has been selected for reliability, scalability, and local maintainability. The phased implementation roadmap ensures that the most critical capabilities are delivered first, building stakeholder confidence and demonstrating value before advancing to more sophisticated features. The budget is designed to be achievable within current fiscal constraints while the subscription model ensures long-term financial sustainability. We respectfully recommend that the Bishoftu City Administration and Oromia Police Commission approve this proposal and allocate the necessary resources to begin Phase 1 implementation at the earliest opportunity. The sooner this system is deployed, the sooner the community will benefit from enhanced safety, improved services, and stronger regulatory oversight of the guest house industry.